

IBM VIT – Agentic AI May 2026

Assignment 1: Smart College Assistant using LangChain Tool Calling Agent

Name : SARATHY P

Reg No : 23MID0094

Batch : IBM Agentic AI -08 (Slot 2)

Email ID : sarathy.p2023@vitstudent.ac.in

Course Title : Agentic AI

QUESTION :

Develop an AI-powered College Assistant using **LangChain Tool** Calling Agent that can automatically identify user requests and invoke the appropriate tool(s) to provide accurate responses.

Requirements

Create the following tools using the `@tool` decorator:

1. Attendance Calculator

Inputs: Total Classes, Attended Classes

Outputs: Attendance Percentage, Exam Eligibility Status

Rule:

- Attendance $\geq 75\%$ $\rightarrow$ Eligible for Exam
- Attendance $< 75\%$ $\rightarrow$ Not Eligible for Exam

2. Result Calculator

Inputs: Marks of 5 Subjects

Outputs: Average Marks, Grade, Pass/Fail Status

AverageGrade

≥ 90 A

75–89 B

60–74 C

< 60 D

Pass if Average ≥ 50 .

3. Fee Balance Calculator

Inputs: Total Course Fee, Amount Paid

Output: Pending Fee Amount

4. Library Fine Calculator

Input: Number of Delayed Days

Output: Fine Amount

Rule: Fine = ₹5 $\times$ Delayed Days

5. Hostel Fee Calculator

Inputs: Monthly Hostel Fee, Number of Months Stayed

Output: Total Hostel Fee

Technical Requirements

Your solution must use:

- Python
- LangChain
- Ollama/OpenAI Model
- @tool
- create_tool_calling_agent()
- AgentExecutor
- ChatPromptTemplate
- verbose=True

Test Cases

Query 1

I attended 72 classes out of 90. Am I eligible for exams?

Query 2

My marks are 95, 90, 88, 91 and 87. What is my grade?

Query 3

My course fee is 50000 and I have paid 35000. How much fee is pending?

Query 4

I returned a library book 8 days late. What is the fine amount?

Query 5

Hostel fee is 6000 per month and I stayed for 5 months. Calculate my hostel fee.

Multi-Tool Challenge

I attended 80 classes out of 100.

My marks are 90, 85, 88, 92 and 95.

My course fee is 60000 and I paid 45000.

Provide:

1. Attendance Status
2. Grade
3. Pending Fee

The agent should automatically invoke multiple tools and provide a consolidated response.

Bonus Challenge

Create a Student Information Tool that retrieves student details from a Python dictionary using Student ID.

SOURCE CODE:

```
from langchain_core.tools import tool
from langchain_core.prompts import ChatPromptTemplate
from langchain.agents import AgentExecutor, create_tool_calling_agent
from langchain_ollama import ChatOllama

# STUDENT DATABASE (BONUS TOOL)

student_db = {
    "ST101": {
        "name": "Lathika",
        "department": "Computer Science",
        "semester": 5
    },
    "ST102": {
        "name": "Pavindhar",
        "department": "Electronics",
        "semester": 3
    },
    "ST103": {
        "name": "Nandana",
        "department": "Information Technology",
        "semester": 7
    }
}

# TOOL 1 – ATTENDANCE CALCULATOR

@tool
def check_attendance(total_classes: int, attended_classes: int) -> dict:
    """
    Calculate attendance percentage and exam eligibility.

    Args:
        total_classes: Total number of classes held.
        attended_classes: Number of classes attended by the student.

    Returns:
        A dict with attendance_percent and exam_status.
    """
    percentage = (attended_classes / total_classes) * 100
    status = "Eligible for Exam" if percentage >= 75 else "Not Eligible for Exam"
    return {
```

```
    "attendance_percent": round(percentage, 2),
    "exam_status": status
}
```

TOOL 2 – RESULT CALCULATOR

@tool

```
def calculate_result(
```

```
    subject1: float,
```

```
    subject2: float,
```

```
    subject3: float,
```

```
    subject4: float,
```

```
    subject5: float
```

```
) -> dict:
```

```
    """
```

Calculate average marks, grade, and pass/fail status for 5 subjects.

Args:

subject1: Marks in subject 1.

subject2: Marks in subject 2.

subject3: Marks in subject 3.

subject4: Marks in subject 4.

subject5: Marks in subject 5.

Returns:

A dict with average_marks, grade, and result_status.

```
    """
```

```
average = (subject1 + subject2 + subject3 + subject4 + subject5) / 5
```

```
if average >= 90:
```

```
    grade = "A"
```

```
elif average >= 75:
```

```
    grade = "B"
```

```
elif average >= 60:
```

```
    grade = "C"
```

```
else:
```

```
    grade = "D"
```

```
result = "PASS" if average >= 50 else "FAIL"
```

```
return {
```

```
    "average_marks": round(average, 2),
```

```
    "grade": grade,
```

```
    "result_status": result
```

```
}
```

TOOL 3 – FEE BALANCE CALCULATOR

@tool

```
def calculate_fee_balance(total_course_fee: float, amount_paid: float) -> dict:
```

```
    """
```

Calculate the pending course fee amount.

Args:

total_course_fee: Total fee for the course.

amount_paid: Amount already paid by the student.

Returns:

A dict with pending_fee.

```
    """
```

```
    pending_fee = total_course_fee - amount_paid
```

```
    return {"pending_fee": pending_fee}
```

TOOL 4 – LIBRARY FINE CALCULATOR

@tool

```
def calculate_library_fine(delayed_days: int) -> dict:
```

```
    """
```

Calculate library fine based on number of delayed days.

Fine = ₹5 × delayed_days.

Args:

delayed_days: Number of days the book was returned late.

Returns:

A dict with fine_amount.

```
    """
```

```
    fine_amount = delayed_days * 5
```

```
    return {"fine_amount": fine_amount}
```

TOOL 5 – HOSTEL FEE CALCULATOR

@tool

```
def calculate_hostel_charges(monthly_hostel_fee: float, months_stayed: int) -> dict:
```

```
    """
```

Calculate total hostel fee based on monthly fee and duration.

Args:

monthly_hostel_fee: Fee charged per month for hostel.
months_stayed: Number of months the student stayed.

Returns:

A dict with total_hostel_fee.

"""

```
total_fee = monthly_hostel_fee * months_stayed
return {"total_hostel_fee": total_fee}
```

BONUS TOOL – STUDENT INFORMATION LOOKUP

@tool

def get_student_record(student_id: str):

"""

Retrieve student information using a Student ID.

Args:

student_id: The unique student identifier (e.g., ST101).

Returns:

Student details dict or a not-found message.

"""

```
if student_id in student_db:
    return student_db[student_id]
return "Student Record Not Found"
```

TOOLS LIST

```
tools = [
    check_attendance,
    calculate_result,
    calculate_fee_balance,
    calculate_library_fine,
    calculate_hostel_charges,
    get_student_record
]
```

OLLAMA MODEL (swap model name as needed)

Other options: "llama3.1", "mistral", "phi3"

```
llm = ChatOllama(
    model="qwen2.5",
    temperature=0
```

```
)
```

```
# PROMPT TEMPLATE
```

```
prompt = ChatPromptTemplate.from_messages([
```

```
(  
    "system",  
    """
```

```
You are SmartCollegeBot, an intelligent assistant for college students.
```

```
Your responsibilities:
```

- Attendance Management
- Result Analysis
- Fee Management
- Hostel Fee Calculation
- Library Fine Calculation
- Student Information Retrieval

```
Instructions:
```

- Carefully understand the student's request.
- If the request needs one tool, call only that tool.
- If the request requires multiple calculations, invoke multiple tools and provide a well-structured summary report.
- Keep responses neat, clear, and professional.
- Always include the ₹ symbol for monetary values.

```
"""
```

```
),  
("human", "{input}"),  
("placeholder", "{agent_scratchpad}")
```

```
])
```

```
# CREATE AGENT & EXECUTOR
```

```
agent = create_tool_calling_agent(
```

```
    llm=llm,  
    tools=tools,  
    prompt=prompt
```

```
)
```

```
agent_executor = AgentExecutor(
```

```
    agent=agent,  
    tools=tools,  
    verbose=True
```

```
)
```



```
# CONSOLE APPLICATION
```

```
print("""
```

```
    SMART COLLEGE ASSISTANT  
    SARATHY P – 23MID0094
```

```
Available Services:
```

1. Attendance Calculator
2. Result Calculator
3. Fee Balance Calculator
4. Library Fine Calculator
5. Hostel Fee Calculator
6. Student Information Lookup

```
Type 'exit' to quit.
```

```
""")
```

```
while True:
```

```
    user_query = input("\nAsk Your Question : ").strip()
```

```
    if not user_query:  
        continue
```

```
    if user_query.lower() == "exit":  
        print("\nThank you for using Smart College Assistant. Goodbye!")  
        break
```

```
try:
```

```
    response = agent_executor.invoke({"input": user_query})  
    print("\n")  
    print("ASSISTANT RESPONSE")  
    print("\n")  
    print(response["output"])  
    print("\n")
```

```
except Exception as e:  
    print(f"\nError: {e}")
```

OUTPUT:

Query 1: I attended 72 classes out of 90. Am I eligible for exams?

```
PS C:\Users\Madava> py -3.11 "C:\Agentic AI\Smart_College_Assistant.py"

SMART COLLEGE ASSISTANT
SARATHY P - 23MID0094

Available Services:
1. Attendance Calculator
2. Result Calculator
3. Fee Balance Calculator
4. Library Fine Calculator
5. Hostel Fee Calculator
6. Student Information Lookup

Type 'exit' to quit.

Ask Your Question : I attended 72 classes out of 90. Am I eligible for exams?

> Entering new AgentExecutor chain...

Invoking: `check_attendance` with `{'total_classes': 90, 'attended_classes': 72}`

{'attendance_percent': 80.0, 'exam_status': 'Eligible for Exam'}Based on your attendance of 72 out of 90 classes, which is 80%, you are eligible to sit for the exams.

> Finished chain.

ASSISTANT RESPONSE

Based on your attendance of 72 out of 90 classes, which is 80%, you are eligible to sit for the exams.

Ask Your Question : exit

Thank you for using Smart College Assistant.
PS C:\Users\Madava>
```

Query 2: My marks are 95, 90, 88, 91 and 87. What is my grade?

```
PS C:\Users\Madava> py -3.11 "C:\Agentic AI\Smart_College_Assistant.py"

SMART COLLEGE ASSISTANT
SARATHY P - 23MID0094

Available Services:
1. Attendance Calculator
2. Result Calculator
3. Fee Balance Calculator
4. Library Fine Calculator
5. Hostel Fee Calculator
6. Student Information Lookup

Type 'exit' to quit.

Ask Your Question : My marks are 95, 90, 88, 91 and 87. What is my grade?

> Entering new AgentExecutor chain...

Invoking: `calculate_result` with `{'subject1': 95, 'subject2': 90, 'subject3': 88, 'subject4': 91, 'subject5': 87}`

{'average_marks': 90.2, 'grade': 'A', 'result_status': 'PASS'}Based on your marks, your average is 90.2 and you have been graded as 'A'. Congratulations! You have passed the course.

> Finished chain.

ASSISTANT RESPONSE

Based on your marks, your average is 90.2 and you have been graded as 'A'. Congratulations! You have passed the course.
```

Query 3: My course fee is 50000 and I have paid 35000. How much fee is pending?

Ask Your Question : My course fee is 50000 and I have paid 35000. How much fee is pending?

> Entering new AgentExecutor chain...

Invoking: `calculate\_fee\_balance` with `{'amount\_paid': 35000, 'total\_course\_fee': 50000}`

{'pending_fee': 15000.0}The pending course fee amount is ₹15000. Please make the payment at your earliest convenience.

> Finished chain.

ASSISTANT RESPONSE

The pending course fee amount is ₹15000. Please make the payment at your earliest convenience.

Query 4: I returned a library book 8 days late. What is the fine amount?

Ask Your Question : I returned a library book 8 days late. What is the fine amount?

> Entering new AgentExecutor chain...

Invoking: `calculate\_library\_fine` with `{'delayed\_days': 8}`

{'fine_amount': 40}The Library fine for returning the book 8 days Late is ₹40.

> Finished chain.

ASSISTANT RESPONSE

The library fine for returning the book 8 days late is ₹40.

Query 5: Hostel fee is 6000 per month and I stayed for 5 months. Calculate my hostel fee.

Ask Your Question : Hostel fee is 6000 per month and I stayed for 5 months. Calculate my hostel fee.

> Entering new AgentExecutor chain...

Invoking: `calculate\_hostel\_charges` with `{'monthly\_hostel\_fee': 6000, 'months\_stayed': 5}`

{'total_hostel_fee': 30000.0}Your total hostel fee for staying 5 months is ₹30,000.

> Finished chain.

ASSISTANT RESPONSE

Your total hostel fee for staying 5 months is ₹30,000.

Multi-Tool Challenge

I attended 80 classes out of 100. My marks are 90, 85, 88, 92 and 95. My course fee is 60000 and I paid 45000.

Provide:

1. Attendance Status 2. Grade 3. Pending Fee

```
PS C:\Users\Madava> py -3.11 "C:\Agentic AI\Smart_College_Assistant.py"
Ask Your Question : I attended 80 classes out of 100. My marks are 90, 85, 88, 92 and 95. My course fee is 60000 and I paid 45000.Provide: 1. Attendance Status 2. Grade 3. Pending Fee

> Entering new AgentExecutor chain...

Invoking: `check_attendance` with `{'total_classes': 100, 'attended_classes': 80}`

{'attendance_percent': 80.0, 'exam_status': 'Eligible for Exam'}
Invoking: `calculate_result` with `{'subject1': 90, 'subject2': 85, 'subject3': 88, 'subject4': 92, 'subjects': 95}`

{'average_marks': 90.0, 'grade': 'A', 'result_status': 'PASS'}
Invoking: `calculate_fee_balance` with `{'total_course_fee': 60000, 'amount_paid': 45000}`

{'pending_fee': 15000.0}Based on the provided information and calculations:

1. **Attendance Status**: Your attendance percentage is 80%, which makes you eligible for the exam.
2. **Grade**: You have an average of 90 marks across your subjects, earning a grade of 'A'.
3. **Pending Fee**: The amount you still need to pay towards your course fee is ₹15,000.

Please ensure that any pending fees are settled promptly to avoid any inconvenience.

> Finished chain.

ASSISTANT RESPONSE

Based on the provided information and calculations:

1. **Attendance Status**: Your attendance percentage is 80%, which makes you eligible for the exam.
2. **Grade**: You have an average of 90 marks across your subjects, earning a grade of 'A'.
3. **Pending Fee**: The amount you still need to pay towards your course fee is ₹15,000.

Please ensure that any pending fees are settled promptly to avoid any inconvenience.
```

Bonus Challenge:

Create a Student Information Tool that retrieves student details from a Python dictionary using Student ID.

```
Ask Your Question : Get details of student ID ST101

> Entering new AgentExecutor chain...

Invoking: `get_student_record` with `{'student_id': 'ST101'}`

{'name': 'Lathika', 'department': 'Computer Science', 'semester': 5}The details of the student with ID ST101 are as follows:
- Name: Lathika
- Department: Computer Science
- Semester: 5

Is there anything else you need help with regarding this information?

> Finished chain.

ASSISTANT RESPONSE

The details of the student with ID ST101 are as follows:
- Name: Lathika
- Department: Computer Science
- Semester: 5

Is there anything else you need help with regarding this information?
```